

Converting Rates to Forecasts

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1 Scoring Forecasts

The goal of forecasting is to guess the value of a variable, which we call a *outcome* and denote by y . To define what makes a good prediction, we need to specify an evaluation metric. Since we're computationally minded people, we tend to evaluate predictions numerically. We thus assume we have a *loss function* that scores a prediction p against a label y , $\text{loss}(p, y)$. There are many different loss functions we might use.

People seem to like evaluating forecasters on their track records, not their greatest hits. This is also the most mathematically convenient way to evaluate forecasts. We can measure some sort of average error over multiple predictions. That is, we'll examine,

$$\frac{1}{n} \sum_{i=1}^n \text{loss}(p_i, y_i) \tag{1}$$

where loss is some appropriately chosen function that scores prediction errors. The canonical example is the square-loss

$$\text{loss}(p, y) = (p - y)^2.$$

The average square loss of a series of predictions is called the *Brier score*. We'll encounter other examples next week. But the Brier score is nice because it works with both binary and real-valued outcomes and lends itself to clean, closed-form calculations that tell us a lot about the culture behind mathematical forecasting.

2 Optimal Forecasts

A weird but useful exercise is to look at what the *optimal forecasts* might be given a particular scoring function. If you know all the outcomes in advance, what would the best predictions be? The cost function ultimately backs us into a particular style of prediction.

Assume we have a series of N binary events to forecast. We know the outcomes $y_1, \dots, y_N$ in advance. Each y_i is either equal to 0 or 1.

Usually we are allowed to make our forecasts using some properties of the event, which we call *attributes* or *features* and denote by x . These are provided to the forecaster before they are asked

to make a prediction. The features are supposed to provide sufficient information for a forecast, so that all individuals with the same feature vector x should be assigned the same prediction (that is, if $x_i = x_j$, then we must have $p_i = p_j$).

With this in mind, we can collect particular statistics about each value of the feature vector. Let q_x denote the fraction of the time that $y_i = 1$ when $x_i = x$. Also, let n_x denote the number of individuals in the data set with $x_i = x$. Let $\mathcal{X}$ denote the set of all possible values of the feature vector x . With this notation, we can collect terms in the average score (Equation (1)) to reveal the optimal forecasts.

$$\begin{aligned} \text{loss}_{\text{pop}} &= \frac{1}{N} \sum_{i=1}^N \text{loss}(f(x_i), y_i) \\ &= \sum_{x \in \mathcal{X}} \frac{n_x}{N} \left\{ \frac{1}{n_x} \sum_{\{i : x_i = x\}} \text{loss}(f(x), y_i) \right\} \\ &= \sum_{x \in \mathcal{X}} \frac{n_x}{N} \{q_x \text{loss}(f(x), 1) + (1 - q_x) \text{loss}(f(x), 0)\} \end{aligned} \quad (2)$$

This final expression is a positive combination of terms that depend on a single x . In particular, this means that the best prediction can be found by individually examining the case of each $x \in \mathcal{X}$. The optimal prediction given the feature vector x is

$$f^*(x) \in \arg \min_p q_x \text{loss}(p, 1) + (1 - q_x) \text{loss}(p, 0).$$

The only information needed to make a prediction is the rates q_x and the definition of the loss function.

In this case of the Brier score, we seek to minimize

$$q_x(p - 1)^2 + (1 - q_x)p^2.$$

with respect to p . The optimal p can be found by equating the derivative with 0:

$$0 = 2q_x(p - 1) + 2(1 - q_x)p = 2p - 2q_x,$$

which means the optimal prediction is just q_x . To make the optimal prediction of y given x under the Brier score, we need only know the frequency that $y = 1$ when the feature vector equals x .

One interesting fact about this calculation is that without constraining our prediction, the optimal forecast in a Brier score must lie in the interval $[0, 1]$. It *feels* like a probability that the event will happen. In the next section, we make this connection even more explicit.

3 From Prediction to Probabilities

A good forecaster wants to have their Brier score as low as possible. A set of forecasts is called *coherent* if there is no set of events where there is another forecast that achieves a lower Brier score no matter the outcome of the events.

Let me give an example of an incoherent forecast: predicting 0.7 for rain tomorrow and 0.4 for no rain.

$$\frac{1}{2}(1 - .7)^2 + \frac{1}{2}(0 - .4)^2 = 0.125$$

If it doesn't rain, the score is

$$\frac{1}{2}(0 - .7)^2 + \frac{1}{2}(1 - .4)^2 = 0.425$$

However, the competing forecast of 0.65 for rain and 0.35 for no rain has Brier score

$$\frac{1}{2}(1 - .65)^2 + \frac{1}{2}(0 - .35)^2 = 0.1225$$

if it rains and

$$\frac{1}{2}(0 - .65)^2 + \frac{1}{2}(1 - .35)^2 = 0.4225$$

if it doesn't rain. These scores are lower for the second forecast than for the incoherent forecast, whether it rains or not.

Interestingly, the predictions submitted in the second forecast add up to 1, whereas those of the incoherent forecast sum to 1.1. What we'll show now is that a forecast is incoherent unless it obeys the axioms of probability. Note that nowhere so far have we assumed that anything is probabilistic. We do not assume that the questions are "generated at random" or "sampled from a "distribution." Instead, it is only the cost function that forces forecasters to operate inside the rules of probability.

The assertion turns out to follow from some really neat properties of Euclidean geometry. Note that the Brier score is nothing more than a Euclidean norm. We can define a vector of predictions $\mathbf{p} = (p_1, \dots, p_n)$ and a vector of outcomes $\mathbf{y} = (y_1, \dots, y_n)$. $\mathbf{y}$ is an element of the n -dimensional hypercube $\{0, 1\}^n$. We have

$$BS := \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2 = \frac{1}{n} \|\mathbf{p} - \mathbf{y}\|^2.$$

Now, the set of valid vectors $\mathbf{y}$ is not all elements of the hypercube. For example, if the first forecast is about whether it will rain and the second about whether it will not rain, then the only two values admissible for those first two components are (0, 1) and (1, 0). The outcomes (0, 0) and (1, 1) cannot occur. We will now show that the predictions must respect these sorts of logical restrictions as well. To do so, we first introduce an important object from geometry, *the convex hull*.

Definition 3.1. Let $\Omega = \{\mathbf{v}_1, \dots, \mathbf{v}_K\}$ be a set of K points in $\mathbb{R}^n$. A *convex combination* of these points is a linear combination where all of the weights are nonnegative and sum to 1. That is, a vector of the form

$$\mathbf{x} = \sum_{k=1}^K q_k \mathbf{v}_k$$

with $q_k \geq 0$ for all k and $\sum_{k=1}^K q_k = 1$.

Definition 3.2. Let $\Omega = \{v_1, \dots, v_K\}$ be a set of K points in $\mathbb{R}^n$. The convex hull of Ω is the set of all convex combinations of Ω .

You can think of the convex hull as filling in the volume contained within a mesh of points. The convex hull is the smallest convex set that contains all of the points. All line segments between two points in the convex hull lie in the convex hull. The convex hull is also the intersection of all half-spaces containing that set of points.

We don't need many properties of convex hulls here, but you can already see that probability plays a role in the definition. The coefficients look a lot like a probability distribution over the points in Ω . The relationship to forecasting is closely connected to this. We'll now show that any forecast that lies outside the convex hull of the possible outcomes $\mathcal{Y}$ is incoherent.

Theorem 3.3. Let $\mathcal{Y} \subset \{0, 1\}^n$ denote the set of all logically possible outcome vectors. Let $\mathbf{p} = (p_1, \dots, p_n)$ be a vector of forecasts that is not in $\text{conv}(\mathcal{Y})$. Then there exists a $\mathbf{p}' \in \text{conv}(\mathcal{Y})$ such that $BS(\mathbf{p}', \mathbf{y}) < BS(\mathbf{p}, \mathbf{y})$ for all $\mathbf{y} \in \mathcal{Y}$.

Let $\Pi_{\mathcal{Y}}(\mathbf{x})$ denote the point in $\text{conv} \mathcal{Y}$ which is closest to Euclidean distance to $\mathbf{x}$. That is $\Pi_{\mathcal{Y}}(\mathbf{x})$ satisfies

$$\|\Pi_{\mathcal{Y}}(\mathbf{x}) - \mathbf{x}\| \leq \|\mathbf{y} - \mathbf{x}\| \quad \text{for all } \mathbf{y} \in \text{conv}(\mathcal{Y}).$$

This point exists and is unique.

Theorem 3.3 is an immediate consequence of the following fact from Euclidean geometry:

Lemma 3.4. For any $\mathbf{x} \in \mathbb{R}^n$

$$\|\mathbf{x} - \mathbf{y}\|^2 \geq \|\Pi_{\mathcal{Y}}(\mathbf{x}) - \mathbf{y}\|^2 + \|\Pi_{\mathcal{Y}}(\mathbf{x}) - \mathbf{x}\|^2$$

for all $\mathbf{y} \in \text{conv} \mathcal{Y}$

The Euclidean projection thus serves as the claimed point in the convex hull with lower Brier Score in Theorem 3.3. If $\mathbf{x}$ is not in the convex hull, the second term will be positive, and hence the projection will achieve a lower Brier score no matter what the future holds. A proof of Lemma 3.4 is at the end of this section.

Now what does this have to do with the laws of probability? As confusing as probability is, it only needs to obey three laws. A probability must be greater than or equal to zero. The probability of an event that is a certainty must equal 1. And the probabilities of two mutually exclusive events must add. That is, if π_A is the probability of event A and π_B is the probability of event B and A and B cannot both be true, then the probability that either A or B occurs is $\pi_A + \pi_B$.

Theorem 3.3 implies that all three must happen. Since all of the vectors in $\mathcal{Y}$ have nonnegative components, every element of their convex hull does too. If the k th forecasting event is certain, then $y_k = 1$ for all $\mathbf{y} \in \mathcal{Y}$. Therefore, a coherent forecast must predict $p_k = 1$.

The additivity condition requires a bit more notation to verify. Consider mutually exclusive events A and B . Suppose without loss of generality that the first prediction is for event A , the

second for event B , and the third is for the prediction of $A \vee B$ (i.e., A or B). The coherent prediction can be written as a convex combination

$$\mathbf{p} = \sum_{\mathbf{y} \in \mathcal{Y}} q_{\mathbf{y}} \mathbf{y}$$

for some nonnegative scalars $q_{\mathbf{y}}$ that sum to 1. But then, letting $\mathcal{Y}_A$ denote the elements of $\mathcal{Y}$ where $y_1 = 1$ and $\mathcal{Y}_B$ where $y_2 = 1$, we have

$$\begin{aligned} p_1 &= \sum_{\mathbf{y} \in \mathcal{Y}_A} q_{\mathbf{y}}, \\ p_2 &= \sum_{\mathbf{y} \in \mathcal{Y}_B} q_{\mathbf{y}}, \\ p_3 &= \sum_{\mathbf{y} \in \mathcal{Y}_A \cup \mathcal{Y}_B} q_{\mathbf{y}} = p_1 + p_2, \end{aligned}$$

proving the additivity condition.

3.1 Proof of Lemma 3.4

By my definition of the convex hull, if $\mathbf{y}_1$ and $\mathbf{y}_2$ are both in the convex hull, then so is the line segment between them. That is, $\mathbf{y}_1 + t(\mathbf{y}_2 - \mathbf{y}_1)$ is in the convex hull for all $t \in [0, 1]$.

Now fix $\mathbf{v} \in \text{conv } \mathcal{Y}$ and let

$$\mathbf{v}_t = \Pi_{\mathcal{Y}}(\mathbf{x}) + t(\mathbf{v} - \Pi_{\mathcal{Y}}(\mathbf{x}))$$

$\mathbf{v}_t \in \text{conv}(\mathcal{Y})$ for all $t \in [0, 1]$. Then since $\Pi_{\mathcal{Y}}(\mathbf{x})$ is the closest point to $\mathbf{x}$ in $\text{conv}(\mathcal{Y})$, we have

$$\begin{aligned} \|\mathbf{x} - \Pi_{\mathcal{Y}}(\mathbf{x})\|^2 &\leq \|\mathbf{x} - \mathbf{v}_t\|^2 \\ &= \|\mathbf{x} - \Pi_{\mathcal{Y}}(\mathbf{x})\|^2 - 2t\langle \mathbf{x} - \Pi_{\mathcal{Y}}(\mathbf{x}), \mathbf{v} - \Pi_{\mathcal{Y}}(\mathbf{x}) \rangle + t^2\|\mathbf{v} - \Pi_{\mathcal{Y}}(\mathbf{x})\|^2. \end{aligned}$$

Since the first term is nonnegative, we have

$$-2t\langle \mathbf{x} - \Pi_{\mathcal{Y}}(\mathbf{x}), \mathbf{v} - \Pi_{\mathcal{Y}}(\mathbf{x}) \rangle + t^2\|\mathbf{v} - \Pi_{\mathcal{Y}}(\mathbf{x})\|^2 \geq 0$$

for all $t \in (0, 1]$. Since this holds for arbitrarily small t , we must have

$$\langle \mathbf{x} - \Pi_{\mathcal{Y}}(\mathbf{x}), \mathbf{v} - \Pi_{\mathcal{Y}}(\mathbf{x}) \rangle \leq 0$$

for all $\mathbf{v} \in \text{conv } \mathcal{Y}$. In particular, this must hold for all $\mathbf{y} \in \mathcal{Y}$.

From this we get the following

$$\|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{x} - \Pi_{\mathcal{Y}}(\mathbf{x})\|^2 - 2\langle \mathbf{x} - \Pi_{\mathcal{Y}}(\mathbf{x}), \mathbf{y} - \Pi_{\mathcal{Y}}(\mathbf{x}) \rangle + \|\Pi_{\mathcal{Y}}(\mathbf{x}) - \mathbf{y}\|^2$$

Since the second term is nonnegative, this proves the lemma.

4 Probabilities as fractions of populations

Probability is unfortunately very confusing and can mean many things depending on context. Probability can just represent fractions of discrete counts. It can represent coherent forecasts. For Bayesians, it can be a quantification of belief. In all cases, probabilities obeys the same three axioms. This makes it easy to casually slip between contexts, turning fractions into forecasts and forecasts into beliefs even if that might not be justified.

I use this warning to preface how it's easy to just start with the idea that the future itself is probabilistic, and optimal forecasts are reduced to calculating conditional probabilities in the probabilistic model of reality. I don't endorse this view, but I present it here because it is a common presentation and it's helpful to be able to connect my non-probabilistic view with the more standard probabilistic one.

So let's return to what we did in Section 2. We can think of the quantity q_x as the conditional probability of y given x under the uniform distribution on the discrete set $\{(x_i, y_i)\}_{i \in I}$.

$$q_x = \Pr[y = 1|x] = \mathbb{E}[y|x]$$

This is the fraction of the time the outcome is equal to 1 when the feature is equal to x .

Similarly, the weighting of each term in the summation of Equation (2) is a probability.

$$\frac{n_x}{N} = \Pr[x]$$

This is the fraction of time the feature is equal to x .

We can combine these two together to define a joint probability distribution.

$$\Pr[x, y] = \Pr[y|x] \Pr[x]$$

This joint probability is the fraction of the time that the feature equals x and the label equals y .

With this set of complementary notation, we can derive a difference view of the population loss:

$$\begin{aligned} \text{loss}_{\text{pop}} &= \frac{1}{N} \sum_{i=1}^N \text{loss}(f(x_i), y_i) \\ &= \sum_{x \in \mathcal{X}} \sum_{y \in \{0,1\}} \Pr[x] \Pr[y|x] \text{loss}(f(x), y) \\ &= \sum_{x \in \mathcal{X}} \sum_{y \in \{0,1\}} \Pr[x, y] \text{loss}(f(x), y) \\ &= \mathbb{E}[\text{loss}(f(x), y)] \end{aligned}$$

That is, the population loss can be thought of as an “expected loss” under our defined probability distribution.

We can define other probabilities as well. If n_y is the number of individuals whose outcome equals y , then

$$\Pr[y] = \frac{n_y}{N}$$

The conditional probability $\Pr[x|y]$ is the fraction of the time that the feature equals x when the label is equal to y . These probabilities often have names in decision theory. $\Pr[y]$ is usually the *prior* on the frequency of outcomes. $\Pr[x|y]$ is called the likelihood. It measures how probable a certain value of the observed feature vector is under the assumption of each outcome. A maximum likelihood prediction would compute the likelihood of x given that $y = 1$ and $y = 0$ and predict the outcome with the higher likelihood.

Since $\Pr[y|x] = \mathbb{E}[y|x]$, this probability is often called the *regression function*. Bayesians will call it the *posterior* when this value is inferred from a prior and likelihood by Bayes' rule.

$$\Pr[y|x] = \frac{\Pr[x|y] \Pr[y]}{\sum_{y' \in \{0,1\}} \Pr[x|y'] \Pr[y']} .$$

We're not going to get into this sort of Bayesian reasoning too much in the class, but I'm leaving it out here so you can be at least passingly familiar if someone tries to snow you with Bayesianism in a debate.